

IEPS: Programming Assignment 3

Explainable Retrieval-Augmented Generation

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I. INTRODUCTION

This assignment extends our PA2 Slovenian information-retrieval system into an explainable Retrieval-Augmented Generation (RAG) pipeline. We retain the successfully evaluated MedOverNet corpus and retrieval configuration, and add answer generation in two directly comparable modes: *With Context*, where retrieved chunks form explicit evidence, and *Without Context*, where the same local language model receives only the question.

The implementation is intentionally experimental rather than deployed as a user interface. For every RAG answer it stores the evidence passages, original vector ranks, reranked ranks and scores, allowing answer errors to be related to retrieval failures rather than treated as opaque model behavior.

II. IMPLEMENTATION SUMMARY

A. Existing PA2 Evidence Store

PA3 reuses the PA2 PostgreSQL/pgvector database without schema changes or re-indexing. The corpus contains Slovenian health and wellness material from MedOverNet, including editorial articles and forum threads. PA2 created `page_segment_long` chunks: articles are grouped into approximately 250 words with 50-word overlap; forum content is kept close to individual posts using approximately 220 words with 30-word overlap.

This was the appropriate continuation of PA2: long segments preserve enough semantic context for a generated response, while PA2 manual evaluation showed that the 50-character comparison chunks frequently lose meaning at chunk boundaries.

B. Models and Libraries

Queries are embedded using `sentence-transformers/LaBSE`, a multilingual sentence embedding model whose 768-dimensional output exactly matches the already populated `vector(768)` columns. Candidate segments are obtained using cosine distance through pgvector and then reranked with `BAAI/bge-reranker-v2-m3`, the multilingual cross-encoder already evaluated in PA2.

Answers are generated through a local Ollama service using `gemma3:4b`. The model has approximately four billion parameters; its Ollama package is approximately 3.3 GB and its documented multilingual support makes it a practical local model for Slovenian questions [1]. A deterministic experimental setting is used with temperature set to zero.

The implementation is divided into:

- `retriever.py`: LaBSE query embedding, cosine retrieval of 20 candidate chunks, and reranking to the final top five evidence chunks;
- `prompts.py` and `generator.py`: separate prompts for evidence-grounded and LLM-only answering, and the Ollama API integration;
- `rag_pipeline.py`: single-question command-line interface;
- `evaluate.py`: batch comparison runner storing full JSON traces and manual scoring fields.

C. Prompt Design

In RAG mode, the prompt includes numbered chunks and source URLs. The model is instructed to use only that evidence, cite used chunks as [1], [2], and explicitly state when the context is insufficient. This is particularly important for medical content and out-of-domain questions.

In LLM-only mode, no chunks or URLs are provided. The model is asked for a short Slovenian response and told not to invent references. Keeping the question and generation model fixed while changing only the evidence enables a transparent comparison.

III. SYSTEM EVALUATION CRITERIA

A. Data Processing

No new data preparation is performed in PA3. The evaluation relies on the PA2 decision to use logical long chunks with overlap. Data-processing quality is therefore assessed through inspectability of the retrieved chunk: a good chunk should contain an interpretable passage that directly supports an answer rather than only a matching keyword.

B. Retrieval

For each query, LaBSE cosine search fetches 20 initial candidates from `page_segment_long`. The cross-encoder reranker chooses five passages supplied to the LLM. We store the vector rank, cosine distance, final rank, reranker score, text and URL for every final passage.

Retrieval relevance is scored manually on a three-point scale: 0 (not informative), 1 (partially informative), and 2 (directly supports answering the question). For each answer we also assess answer quality, groundedness in the visible passages, and whether the evidence makes the result diagnosable.

C. Failure Categories

Failures are separated into four types: irrelevant retrieval, insufficient corpus coverage, ambiguous query, and generation hallucination. This distinction prevents attributing an incorrect answer to the generator when the index never contained suitable evidence.

IV. EVALUATION QUERY SET

Table I lists the nine controlled evaluation questions. The first three good queries preserve PA2 comparability. The additional topics were selected only after they were observed in stored PA2 extracted content or PA2 retrieval outputs. The three adverse cases are retained from PA2 because they expose ambiguity and corpus-boundary failures.

Table I
PA3 CONTROLLED QUERY SET AND INTENDED DIAGNOSTIC ROLE

Type	Query	Coverage basis / expected behavior
Good	Kateri so simptomi visokega krvnega tlaka in zakaj je merjenje pomembno?	PA2 reranking surfaced a direct hypertension-management article.
Good	Kako zdrav življenjski slog pomaga pri sladkorni bolezni tipa 2?	PA2 surfaced a diabetes and cardiometabolic lifestyle article.
Good	Kako izboljšati spanec in zakaj je reden spanec pomemben?	PA2 retrieval and validation content contain sleep-focused articles.
Good	Kateri so značilni simptomi Parkinsonove bolezni in kako pomaga gibanje?	The extracted PA2 validation sample contains a Parkinson disease article.
Good	Kateri znaki lahko kažejo na pomanjkanje vitamina B12?	The extracted PA2 validation sample contains a vitamin B12 article.
Good	Zakaj so beljakovine pomembne pri ohranjanju mišične mase?	A saved PA2 retrieval run contains protein and muscle-preservation passages.
Bad	Boli me	Underspecified symptom; retrieval cannot establish the user’s intended pain.
Bad	Kakšne so pravne posledice neplačila prometne kazni v Sloveniji?	Legal question outside the medical corpus; RAG should decline evidence-based answering.
Bad	ATM	Unspecified acronym; PA2 already showed unrelated retrievals and very low reranker scores.

V. EVALUATION RESULTS WORKFLOW

The executable evaluation runner writes a JSON record for each question with: the top-five reranked chunks, both generated answers, exact model and retrieval parameters, and manual assessment fields. This design avoids silently selecting only favorable answers and makes the comparison reproducible.

At the time this report artifact was built, generation results were not fabricated: the local verification environment did not provide a running Ollama service or active PA2 database. Before submission, the fixed experiment is run with:

```
python pa3/rag/evaluate.py
```

The generated `pa3/rag/runs/<timestamp>_evaluation.json` is then the source of truth for the final comparison table and manual scores. Table II specifies the values recorded for each row, ensuring that completing the report is mechanical rather than subjective reimplementations of the experiment.

Table II
PER-QUERY EVIDENCE RECORDED BY THE EVALUATION RUNNER

Field	Purpose
Retrieved context	Inspect evidence and URLs presented to RAG.
RAG answer	Determine grounding and explicit citation use.
LLM-only answer	Compare fluency and unsupported specificity.
Relevance (0–2)	Judge retrieval success.
Quality (0–2)	Compare both answer modes.
Failure type	Locate retrieval, coverage, ambiguity, or hallucination error.

VI. EXPECTED ANALYSIS AND LIMITATIONS

Based on the PA2 evidence, well-specified medical questions should benefit from RAG because relevant Slovenian chunks are available and citations let a reader inspect the basis of an answer. In contrast, the legal query cannot be answered from this corpus. A correct RAG behavior is therefore a transparent

refusal to infer legal consequences from irrelevant medical evidence, even if the LLM-only response sounds plausible.

The “Boli me” and “ATM” cases test a different limitation: the corpus may contain health content, but the question does not identify a useful intent. Reranking can reorder candidates, but it cannot resolve missing user context. The prompt is therefore designed to favor an insufficiency statement over a confident unsupported answer.

The primary experimental limitation is qualitative assessment: nine questions provide interpretable case analysis rather than a statistically representative benchmark. This is appropriate to the assignment’s focus on transparency, but it should not be interpreted as clinical performance evaluation.

VII. CONCLUSION

PA3 extends the strong PA2 retrieval system without disrupting its documented choices. It adds a local Ollama generation layer, an evidence-restricted RAG prompt, an LLM-only comparison mode, and complete retrieval traces for evaluation. The central explainability benefit is concrete: every RAG answer can be checked against the passages that shaped it, and failures can be classified as missing evidence, poor retrieval, ambiguous input, or model generation error.

REFERENCES

- [1] Ollama, “gemma3 model library,” <https://ollama.com/library/gemma3>, accessed May 25, 2026.